

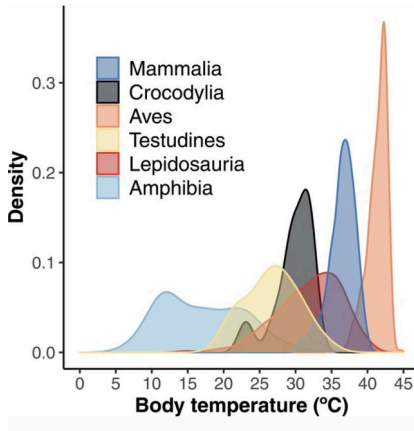


Physiology

Dr. Ethan Linck
Office: 305 Lewis Hall

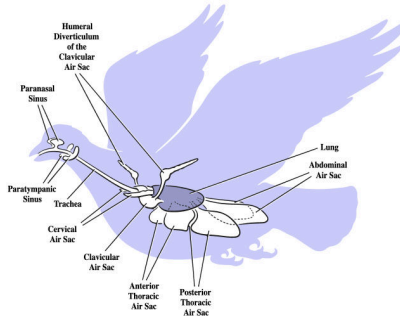
High bird body temperatures aided adaptive radiation

- ▶ Birds maintain body temperature of 40° C through *endothermy*—metabolic production of heat
- ▶ Higher than all other reptiles, mammals, and amphibians



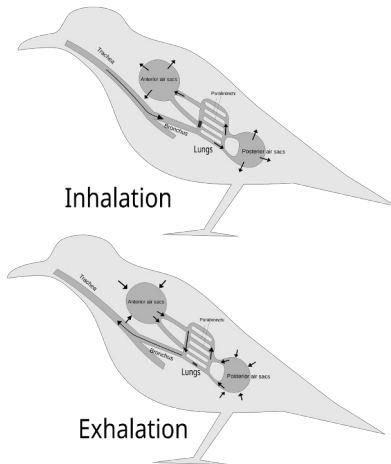
Air sacs facilitate respiration

- ▶ Thin-walled, non-vascular compartments throughout body cavity and cervical vertebrae
- ▶ 6 (in weavers) to 12 (shorebirds and storks)
- ▶ Connected to lungs by **secondary bronchi**



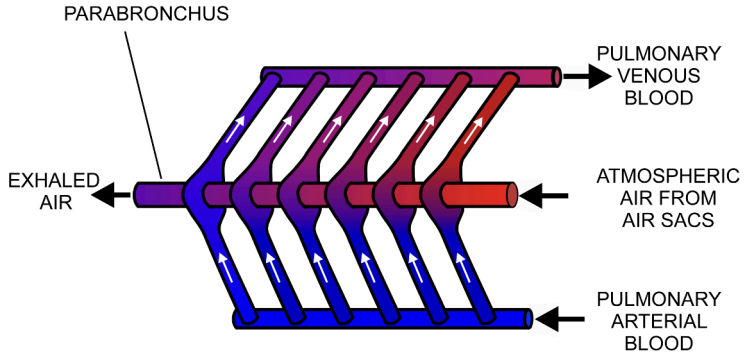
Air sacs facilitate respiration

- ▶ Air sacs expand and contract with entire abdominal cavity
- ▶ Forces air through rigid lungs



Unidirectional airflow in bird lungs maximizes O_2 transfer

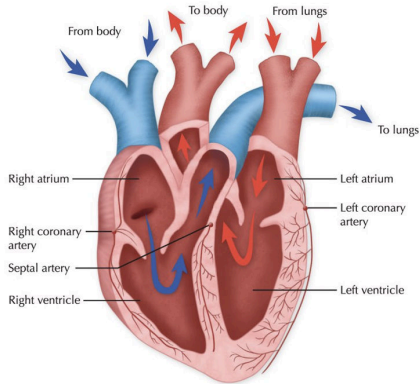
- ▶ No residual air left in lungs during ventilation
- ▶ More efficient gas exchange



(Wikipedia)

Birds have a four-chambered heart

- ▶ 2-4% of body mass (versus $<1\%$ in small mammals)
- ▶ Typically 150-350 bpm (1200 bpm in hummingbirds)
- ▶ Lower resting heart rate than similarly sized mammals
- ▶ High blood pressure!



Thermoregulation

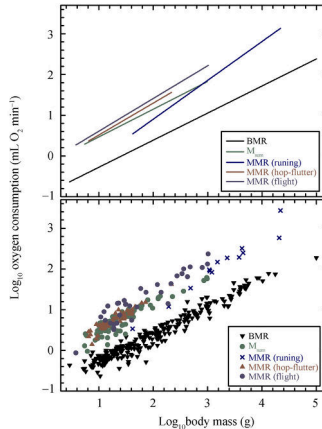
- ▶ Endotherms must cope with metabolic heat in hot environments and heat loss in cold environments
- ▶ Can make physical adjustments, physiological changes, and behavior



Pterocles alchata

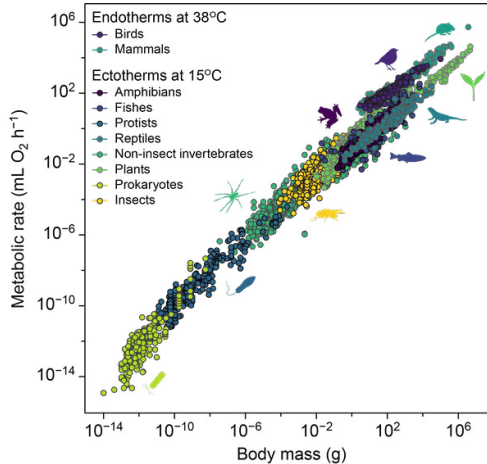
Metabolism

- ▶ **Basal metabolic rate** / BMR: energy expenditure when resting
- ▶ **Active metabolic rate**: energy expenditure when active



Birds have high basal metabolic rate

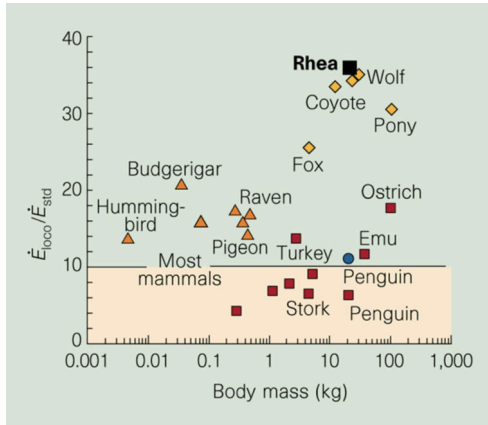
- High relative to most vertebrates



(Tommy Norin)

Birds have high metabolic scope

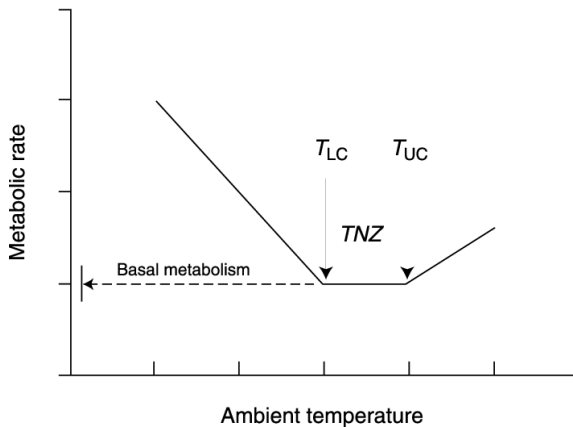
- ▶ **Metabolic scope** = active metabolism / resting metabolism
- ▶ Especially high in running birds



(Bundle et al. 1999)

Low energy expenditure for temperature regulation in thermoneutral zone

- ▶ T_{LC} : Lower critical temperature
- ▶ T_{UC} : Upper critical temperature

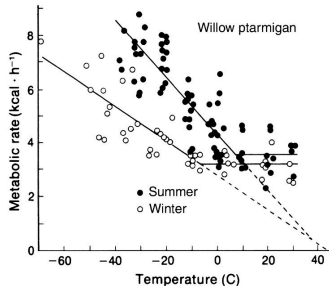


Which statement is most correct?

- a) BMR can be measured in a resting bird below T_{LC}
- b) BMR can be measured in an active bird above T_{LC} and below T_{UC}
- c) BMR can be measured in a resting bird above T_{LC} and below T_{UC}
- d) BMR can be measured in a resting bird above T_{UC}
- e) BMR can be measured in an active bird below T_{LC}

Plumage impacts metabolic temperature regulation

- ▶ Summer plumage less insulative than winter plumage in Willow Ptarmigan, requires higher BMR for a given temperature



(ML93964071; Speakman et al. 2010)

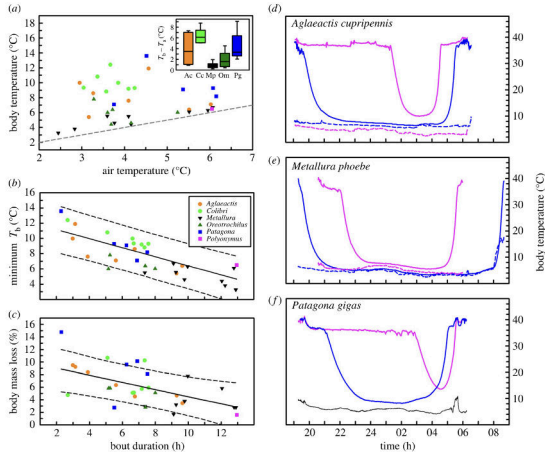
Hypothermia

- ▶ **Nocturnal hypothermia:** T_{body} drops 8-10° C to conserve energy (doves, pigeons, vultures, some passerines)
- ▶ **Torpor:** T_{body} drops >10° C, reduce respiratory rate (1-2x per minute versus 300-400 in hummingbirds), reduce heart rate (30 bpm versus 500), reduce energy consumption 10-60%



Torpor helps Andean hummingbirds cope with altitude

- ▶ Blue and purple lines individual T_{body} measurements
- ▶ Dashed blue line ambient temperature



Why is the duration of torpor bout inversely correlated with body mass loss?

- a) birds in torpor burn fewer calories
- b) birds in torpor lose less water to the atmosphere through respiration
- c) birds in torpor have greater metabolic scope
- d) a & b
- e) a, b, & c

Hibernation

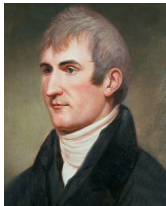
- ▶ Common Poorwills drop temperature from 106° F to 65° F for 2-3 months
- ▶ “Discovered” in 1946, but Hopi Indian name (*Hölchko*) means “The Sleeping One”!



Phalaenoptilus nuttallii

Meriwether Lewis on poorwills (1806)

It appeared to be passing into the dormant state... the bird could scarcely move. I run [sic] my penknife into its body under the wing and completely destroyed its lungs and heart yet it lived upwards of two hours. This fanominon [sic] I could not account for unless it proceeded from want of circulation of the blood



Hyperthermia

- Birds can reduce T_{body} by respiration: **panting** (evaporative cooling of nasal, buccal, pharynx regions) and **gular flutter** (special pant via vibrating hyoid muscles / bones in throat; targets mouth and upper respiratory tract)

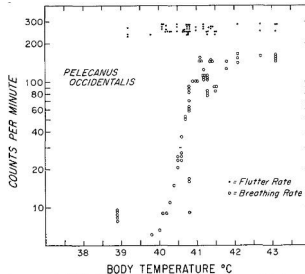
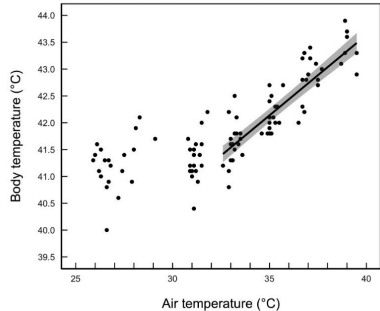


FIGURE 1. The relation of rates of gular flutter and breathing to body (cloacal) temperature in a Brown Pelican weighing 3.13 kilograms.

https://www.youtube.com/watch?v=3_Ft32uEuaA

Physiology informs climate change predictions

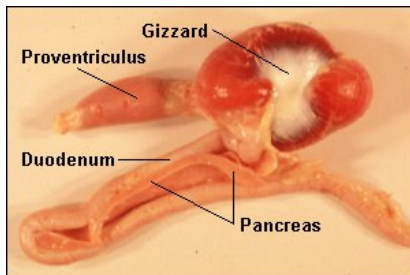
- ▶ Snow Buntings have high cold tolerance at cost of reduced heat tolerance



(ML63891021; O'Connor et al. 2021)

Avian digestion

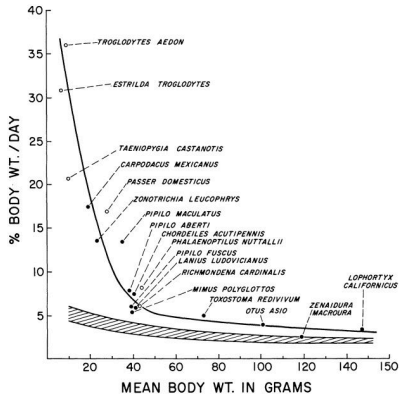
- ▶ **Proventriculus** secretes mucus, HCl + pepsinogen (converted to protein-digesting **pepsin**) to chemically digest food
- ▶ **Gizzard** (ventriculus): muscular avian equivalent of molars, with keratinized lining + sometimes grit
- ▶ **Crop**: expanded esophagus chamber that stores and softens food + regulates flow



(Colorado State)

Water balance

- ▶ Evaporative water loss high per unit mass in small birds
- ▶ Partially offset by **metabolic water** production (enhanced by high avian metabolisms)



(Bartholomew & Cade 1963)

Excretion

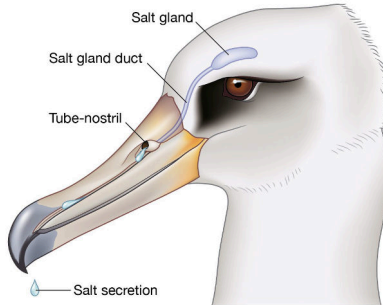
- ▶ Kidneys concentrate nitrogenous wastes as **uric acid** (0.5-1 mL H_2O to excrete 370 mL N versus 20 mL in mammals)



(Natural History Museum)

Avian osmoregulation

- ▶ Birds lack loops of Henle used by mammalian kidneys for salt excretion
- ▶ Seabirds drink seawater (3% salt versus 1% salt in body fluids), must excrete through **nasal salt glands**



References

- ▶ Title slide species: *Pipreola frontalis*, *Nothoprocta curvirostris*, *Dryobates stricklandi*, *Ptilinopus jambu*
- ▶ Moreira, M. O., Qu, Y. F., & Wiens, J. J. (2021). Large-scale evolution of body temperatures in land vertebrates. *Evolution Letters*, 5(5), 484-494.
- ▶ Gill, F. B., Prum, R. O., & Robinson, S. K. (2019). *Ornithology* (4th ed.). W. H. Freeman.
- ▶ McKechnie, A. E., & Swanson, D. L. (2010). Sources and significance of variation in basal, summit and maximal metabolic rates in birds. *Current Zoology*, 56(6), 741-758.
- ▶ Bundle, M. W., Hoppeler, H., Vock, R., Tester, J. M., & Weyand, P. G. (1999). High metabolic rates in running birds. *Nature*, 397(6714), 31-32.

References

- ▶ Bozinovic, F., Ferri-Yáñez, F., Naya, H., Araújo, M. B., & Naya, D. E. (2014). Thermal tolerances in rodents: species that evolved in cold climates exhibit a wider thermoneutral zone. *Evolutionary Ecology Research*, 16(2), 143-152.
- ▶ Speakman, J. R., & Król, E. (2010). Maximal heat dissipation capacity and hyperthermia risk: neglected key factors in the ecology of endotherms. *Journal of Animal Ecology*, 79(4), 726-746.
- ▶ Wolf, B. O., McKechnie, A. E., Schmitt, C. J., Czenze, Z. J., Johnson, A. B., & Witt, C. C. (2020). Extreme and variable torpor among high-elevation Andean hummingbird species. *Biology Letters*, 16(9), 20200428.
- ▶ Bartholomew, G. A., Lasiewski, R. C., & Crawford, E. C. (1968). Patterns of panting and gular flutter in cormorants, pelicans, owls, and doves. *The Condor*, 70(1), 31-34.

References

- ▶ O'Connor, R. S., Le Pogam, A., Young, K. G., Robitaille, F., Choy, E. S., Love, O. P., . . . & Vézina, F. (2021). Limited heat tolerance in an Arctic passerine: Thermoregulatory implications for cold-specialized birds in a rapidly warming world. *Ecology and evolution*, 11(4), 1609-1619.
- ▶ Bartholomew, G. A., & Cade, T. J. (1963). The water economy of land birds. *The Auk*, 80(4), 504-539.